Diffusion: Generation of MNIST digists 3 Möjliga Poäng

5/4/2025

Försök 1

Pågår

NÄSTA: Lämna in uppgift

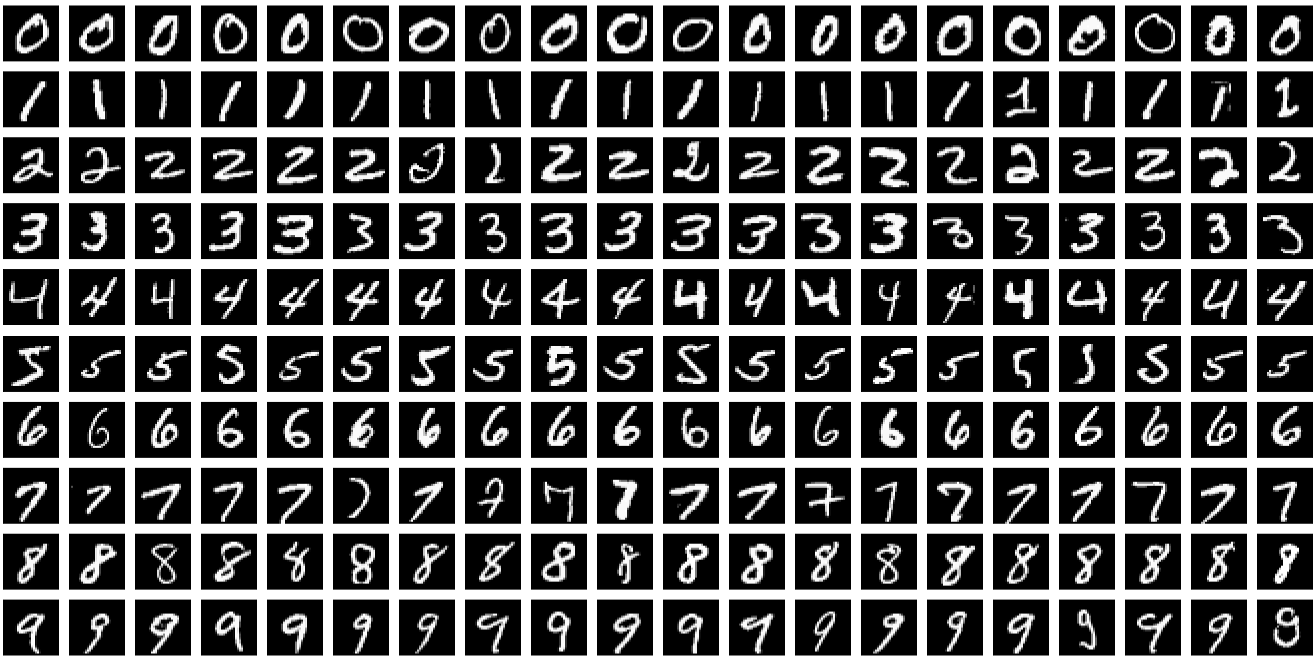
Lägg till kommentar

Obegränsat antal försök tillåts

5/5/2025

Information

In this assignment, you will design and train a Denoising Diffusion Probabilistic Model (DDPM) to generate handwritten digits using the MNIST dataset. The MNIST dataset, which you are already familiar with from the previous course, is a large collection of handwritten digits commonly used for training image processing systems. Below are a few examples from the dataset.



The dataset can be conveniently loaded directly through torchvision via

`import torchvision.datasets as datasets`

`dataset = datasets.MNIST(root="dataset/", download=True)`

You can use the library https://github.com/lucidrains/denoising-diffusion-pytorch (https://github.com/lucidrains/denoising-diffusion-pytorch) which you can install with `pip install denoising_diffusion_pytorch`

Use a U-Net architecture for the DDPM.

You can start from this template: **A05MNIST_diffusion_template.py** (https://uppsala.instructure.com/courses/102453/files/8450901?wrap=1) ttps://uppsala.instructure.com/courses/102453/files/8450901/download?download_frd=1)

Hint: The best normalization for this problem is to keep the color amplitudes between 0 and 1.

Warning: Training a diffusion network can be pretty time-consuming. It is fine if you can't train the diffusion network until convergence.

Remember always to include:

- A **written summary** (0.5–1 A4 page) covering (submitted either as PDF or directly as text):
  - What you did and how
  - What results you obtained
  - What challenges you encountered and what could be improved
- A **PDF (or similar format)** with all **result plots**, each with a short explanation
- Your **code**, preferably as a link (e.g., GitHub, Google Colab, etc.) so we can view it easily.

! Kom ihåg att den här inlämningen räknas för alla i din PhD exercise group grupp.

Välj en inlämningstyp

Text | Ladda upp | Canvas Studio | Mer

< (https://uppsala.instructure.com/courses/102453/modules/items/1358848)

Skicka uppgift